

Subhajit Gorai

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EDUCATION

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech in Information Technology, Currently in Final year – CGPA: 8.01/10

Aug 2023 – June 2027

Shibpur, West Bengal, IN

EXPERIENCE

Amazon

Jan 2027 – June 2027

Incoming Software Development Engineer Intern (SDE)

On-site

Chennai Mathematical Institute (CMI)

May 2026 – August 2026

Research Intern, Machine Learning — Instructor: Prof. Pranabendu Misra

Hybrid

- Engineered a contrastive fine-tuning pipeline for EmbeddingGemma-300M, replacing cosine similarity with the **Angle score** to resolve **InfoNCE**'s gradient saturation
- Extended the pipeline with a τ^+ – **debiased denominator correction** and **adaptive angular margins** on synthesized negation pairs, mitigating false-negative bias in the contrastive loss
- Benchmarked embedding pooling strategies on the **STS-B** benchmark using Gemma-3-1B-PT, achieving 0.5564 **Spearman correlation** with mean pooling

TEXMiN – IIT(ISM) Dhanbad

Feb 2026 – October 2026

Research Fellow, AI-ML & CPS — NM-ICPS, Govt. of India

Remote

- Architected a three-tier zonal system: sensor-layer data acquisition (Zone A), Raspberry Pi edge nodes running **LSTM models** for zone-local anomaly detection and instant alerting (Zone B), and a central aggregation server for mine-wide analysis (Zone C)
- Applied **time-series forecast** and edge ML deployment to build a fault-tolerant CPS pipeline for a safety-critical environment
- Secured an INR 1 Lakh research grant from the National Mission on Interdisciplinary Cyber-Physical Systems to build an AI/ML powered CPS for real-time hazard prediction in underground coal mines

Accenture

May 2026 – July 2026

Advanced Application Engineering Analyst Intern (AEH)

Kolkata, West Bengal, IN

- Designed and deployed an AI onboarding agent in Microsoft Copilot Studio that retrieves and cross-references MS Teams chats and internal PDFs to surface exact room and building numbers for new joiners instead of vague instructions
- Converted internship into a full-time Pre-Placement Offer (PPO) based on performance

PROJECTS

Computable Poetry | [Documentation](#) | *Formal Grammars, Graph Theory, Natural Language Processing* Dec 2025 – June 2026

- Built an ML-free Bangla poem generator using a **two-level CFG** (semantic grammar over POS grammar) with randomized recursive descent and **constraint satisfaction** under strict priority ordering (mātrā > semantic tag > POS > rhyme class)
- Reduced cross-line semantic coherence to a **k-hop shortest path problem** on a layered directed graph; ran Dijkstra over formally derived edge weights to compute the globally optimal thematic trajectory
- Constructed an **inverted index** over a tagged lexicon scraped from 300+ classical poems, keyed by semantic tag and mātrā for **O(1)** candidate lookup, with rhyme equivalence classes, hierarchical fallback, and backtracking on empty sets

OpenCanvas | [Live](#) | [GitHub](#) | *MERN Stack | Google OAuth*

Jan 2025 – Feb 2026

- Sustained **~3,000 RPS at 16ms avg latency** (p99 32ms) querying directly against a 1.2M+ document MongoDB collection—verified via isolated load testing with caching disabled—by replacing offset pagination with cursor-based keyset indexing, eliminating Mongoose hydration overhead via lean queries, and denormalizing snapshots
- Achieved **zero failures across 8,600 simulated requests** and doubled single-thread peak throughput to **6,138 RPS** (15ms median latency) by layering a self-built in-memory TTL cache—featuring lazy/background eviction and a stampede lock—over complex database queries

TECHNICAL SKILLS

Languages: Python, C & C++, JavaScript, TypeScript, Java, SQL

Web Development: Node.js, Express.js, Flask, MongoDB, React, Next.js, HTML, CSS, Tailwind CSS

Machine Learning: PyTorch, NumPy, Pandas, Matplotlib

Platforms: Linux, Git & GitHub

ACHIEVEMENTS

CodeForces: Expert, Max Rating **2029** (Candidate Master) | [Profile](#)

CodeChef: 5-stars, Max Rating **2026** | Secured Global Rank 31 in Starters 239 (Div. 2) | [Profile](#)

Web Development Lead: Development lead at **CodeIIEST (Official Coding Club of IEST Shibpur)**;

Co-developed the official Student Senate portal of our college improving communication for **4,000+** students

Hacktoberfest 2024: Merged 5+ PRs in open source repositories and completed the event

IISER Aptitude Test (IAT): Secured **AIR 453** (CRL)

KVPY SA: Secured **AIR 2063** (CRL)